Response to Reviewers

Paper: A Unified Analysis of Generalization and Sample Complexity for Semi-Supervised Domain Adaptation

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Reviewer 1, Comment 1

Comment: *Below Theorem 1, there is a claim that "This statement holds with probability approaching 1 at an exponential rate with the increase in number of labeled samples M_s", which seems to be unrigorous. Since the second term w.r.t. the target domain in Eq. (5) would not vanish unless $\alpha = 0$, the term seems to serve as a constant. Detailed justifications are expected.*

Response: No action needed.

Reviewer 1, Comment 2

Comment: *The setting of the kernel map in Section 3.2 looks strange. Specifically, if the mappings f^s and f^t for the shared representation learning are both set as ϕ, how could ϕ ensure that the original spaces $\mathcal{X}^s$ and $\mathcal{X}^t$ are projected into the same space? Moreover, in such a setting, this theory may be limited to the scenarios that the source space and target space share the same or similar structures, e.g., it would fail in the heterogeneous cases.*

Response: Reviewer Section 3.2 demiş ama sanırım 2.3'ü kastediyor. Kafa karışıklığını önlemek için Section 2.3'teki kernel map discussion'u, Remark 2 ile birleştirilerek footnote olarak yazıldı. Bu sayede $f^s = f^t$ varsayımının $\mathcal{X}^s = \mathcal{X}^t$ varsayımından geldiği; ama bizim böyle bir varsayımı yine de kullanmadığımız açıkça ifade edilmiş oldu. Bonus olarak yer kazanmış olduk.

Revisions in Manuscript:

↪ **Section 2, Page 8:**
KMM methods set the source and target mappings $f^s : \mathcal{X}^s \to \mathcal{X}$ and $f^t : \mathcal{X}^t \to \mathcal{X}$ as a kernel-induced feature map ϕ. The source and target domains $\mathcal{X}^s = \mathcal{X}^t$ are often assumed to be the same and the transformations are set as $f^s = f^t = \phi$. The shared domain $\mathcal{X}$ is typically a Hilbert space with a kernel $k : \mathcal{X}^s \times \mathcal{X}^t \to R$ satisfying $k(x^s, x^t) = \langle \phi(x^s), \phi(x^t) \rangle_{\mathcal{X}}$ with respect to the inner product $\langle \cdot, \cdot \rangle_{\mathcal{X}}$ in $\mathcal{X}$. However, we do not make these assumptions. We only assume that the distribution discrepancy between ν_s and ν_t is taken as in (2.6) and the empirical estimate is computed as in (2.7).

Reviewer 1, Comment 3

Comment: *The finite variance/moment assumption, i.e., Assumption 3, needs further clarification. In the current version, there is a simple justification that "these assumptions hold for many common data distributions in practice". Since this assumption is important, it would be helpful to provide a more informative discussion, e.g., which types of data would satisfy this condition, or what conditions are necessary for the data?*

Response: Hocam yanlış anlamadıysam bounded input ve neural networkler için bu varsayım direkt geçerli. Bunları da zaten Assumption 5'te ve Lemma 7'de gösteriyoruz. Aşağıdaki açıklamaları da "These assumptions hold for many common data distributions in practice" cümlesine bir footnote olarak ekledim:

Revisions in Manuscript:

↪ **Section 4, Page 9:**
While Assumption 2.5 is stated as a general moment condition, it is naturally satisfied in the context of deep neural networks with bounded input domains. As formally shown later in our analysis of covering numbers (Lemma SM1.7), standard neural networks map compact input domains (such as images with bounded pixel values, as defined in Assumption 3.1) to bounded feature representations. Since any bounded random variable satisfies the Bernstein moment growth condition, Assumption 2.5 holds by definition for the domain-adaptive neural networks studied in Section 3 and the datasets (MNIST, MNIST-M) used in Section 4.

Reviewer 1, Comment 4

Comment: *Following the last question, if the functions f^s and f^t are related to the RKHSs, it would also be helpful to clarify the connection between the assumption and the kernels. Specifically, the finite assumption on kernel-based statistics would be closely related to the choices of kernels.*

Response: Belki Comment 1.3 bu doğrultuda improve edilebilir fakat ben bu comment'e özel bir aksiyon almadım.

Reviewer 1, Comment 5

Comment: *The convergence bound of empirical MMD seems to have already been studied in existing literature. It would be helpful to discuss the differences between the results in Lemma 3 and existing bounds.*

Response: Hocam sanırım Gretton et al. (2012) A Kernel Two-Sample Test makalesinde bazı boundlar var. Üzerinde detaylı düşünemedim ve makalede bu konuyla ilgili bir değişiklik yapmadım fakat bilginize sunarım (ekran alıntısı adı geçen makaleden):

4.1 Bound on the Biased Statistic and Test

We establish two properties of the MMD, from which we derive a hypothesis test. First, we show that regardless of whether or not $p = q$, the empirical MMD converges in probability at rate $O((m+n)^{-\frac{1}{2}})$ to its population value. This shows the consistency of statistical tests based on the MMD. Second, we give probabilistic bounds for large deviations of the empirical MMD in the case $p = q$. These bounds lead directly to a threshold for our first hypothesis test. We begin by establishing the convergence of $\mathrm{MMD}_b[\mathcal{F}, X, Y]$ to $\mathrm{MMD}[\mathcal{F}, p, q]$. The following theorem is proved in A.2.

Theorem 7 *Let p, q, X, Y be defined as in Problem 1, and assume $0 \le k(x,y) \le K$. Then*

$$\Pr_{X,Y}\left\{|\mathrm{MMD}_b[\mathcal{F},X,Y] - \mathrm{MMD}[\mathcal{F},p,q]| > 2\left((K/m)^{\frac{1}{2}} + (K/n)^{\frac{1}{2}}\right) + \varepsilon\right\} \le 2\exp\left(\frac{-\varepsilon^2 mn}{2K(m+n)}\right),$$

where $\Pr_{X,Y}$ denotes the probability over the m-sample X and n-sample Y.

Our next goal is to refine this result in a way that allows us to define a test threshold under the null hypothesis $p = q$. Under this circumstance, the constants in the exponent are slightly improved. The following theorem is proved in Appendix A.3.

Theorem 8 *Under the conditions of Theorem 7 where additionally $p = q$ and $m = n$,*

$$\mathrm{MMD}_b[\mathcal{F},X,Y] \le \underbrace{m^{-\frac{1}{2}}\sqrt{2\mathbf{E}_{x,x'}\left[k(x,x) - k(x,x')\right]}}_{B_1(\mathcal{F},p)} + \varepsilon \le \underbrace{(2K/m)^{1/2}}_{B_2(\mathcal{F},p)} + \varepsilon,$$

both with probability at least $1 - \exp\left(-\frac{\varepsilon^2 m}{4K}\right)$.

In this theorem, we illustrate two possible bounds $B_1(\mathcal{F}, p)$ and $B_2(\mathcal{F}, p)$ on the bias in the empirical estimate (5). The first inequality is interesting inasmuch as it provides a link between the bias bound $B_1(\mathcal{F}, p)$ and kernel size (for instance, if we were to use a Gaussian kernel with large σ, then $k(x,x)$ and $k(x,x')$ would likely be close, and the bias small). In the context of testing, however, we would need to provide an additional bound to show convergence of an empirical estimate of $B_1(\mathcal{F}, p)$ to its population equivalent. Thus, in the following test for $p = q$ based on Theorem 8, we use $B_2(\mathcal{F}, p)$ to bound the bias.[10]

Corollary 9 *A hypothesis test of level α for the null hypothesis $p = q$, that is, for $\mathrm{MMD}[\mathcal{F}, p, q] = 0$, has the acceptance region* $\mathrm{MMD}_b[\mathcal{F},X,Y] < \sqrt{2K/m}\left(1 + \sqrt{2\log\alpha^{-1}}\right)$.

Reviewer 1, Comment 6-7

Comment: *As far as I understand this submission, the major results are the generalization bound in Theorem 1 and its connection to specific algorithms that characterize the discrepancy $D(f^s, f^t)$. However, an important prior is Assumption 1, which can be taken as a stronger version of the change of measure inequality. Note that the L.H.S. of Eq. (3) is dominated by both variables X and Y, while the R.H.S. (i.e., $D(f^s, f^t)$) seems to only depend on X. Such scenarios make this assumption less informative or too strict. More discussion would be helpful.*

Comment: *Following the last question, the analysis on sample complexity under popular learning algorithms is interesting, but the constant R is also important, in my understanding. It seems that the constant R is unjustified throughout this manuscript. It would be expected to show the nature of R explicitly, e.g., how MMD-based network and domain adversarial network satisfy Assumption 1 and what are their corresponding values of R.*

Response: Bu konuya eğileceğiz (Assumption 1).

Reviewer 1, Comment 8

Comment: *The result $\alpha_{optimal} = O(\sqrt{M_t})$ is consistent with the empirical observation, which is good. However, since α should be set within $(0, 1)$, is there any further suggestion for a practical setting for it?*

Response: No action needed.

Reviewer 2, Comment 1

Comment: *The paper relies on assumptions that may not hold in practical domain adaptation scenarios. For example, The theoretical framework relies on Assumption 1, which posits a linear bound between the loss difference and distribution discrepancy (e.g., |Ls - Lt|<=RD(fs,ft)). This assumption may not hold in complex real-world scenarios where domain shifts involve non-linear or high-dimensional disparities, such as label skew or heterogeneous data structures. For instance, the paper acknowledges that domain adaptation must handle "covariate shift, label shift, and heterogeneous settings", yet the assumption simplifies these into a uniform bound. Strengthening this by incorporating more flexible discrepancy measures (e.g., conditional distributions) could improve robustness.*

Response: Bu konuya eğileceğiz (Assumption 1).

Reviewer 2, Comment 2

Comment: *The sample complexity bounds derived for neural networks (e.g., quadratic scaling with width and depth) are not proven to be tight. The paper acknowledges that these bounds are pessimistic but does not compare them to empirical scaling or provide lower bounds, limiting their practical utility. For example, the sample complexity bounds depend heavily on covering numbers of function classes. However, for deep neural networks, covering numbers are often computationally intractable to estimate and can lead to loose bounds that scale quadratically with depth/width ($O(d^2L^2)$).*

Response: No action needed.

Reviewer 2, Comment 3

Comment: *Assumption 5 restricts network parameters to a bounded set. While this simplifies analysis, it contradicts common practices where parameters are regularized but not strictly bounded. In real training, unbounded optimization (e.g., via gradient descent) could violate this assumption, undermining the generality of the bounds. Incorporating norm-based regularization explicitly into the theory would align better with empirical practices.*

Response: No action needed.

Reviewer 2, Comment 4

Comment: *The experiments deliberately maintain networks in the overfitting regime to study sample complexity. While this isolates the effect of network size, it ignores practical techniques like dropout or early stopping that mitigate overfitting. Consequently, the reported sample complexities (e.g., $O(d^2L^2)$) may be overstated for real-world applications. Including ablation studies with regularization could provide a balanced view.*

Response: Hocam ana metinden "deliberately kept in overfitting regime" ifadesini kaldırdım, ve deneylerle ilgili detay verdiğimiz footnote'lara aşağıdaki ifadeleri açıkça ekledim. Bunları daha da öne çıkarıp, bir remark olarak mesela, ifade etmek de mümkün tabii. Yani teorik analizimizde dahil etmedik ama deneysel olarak gösterdik ki bulgularımız gerçek hayat pratikleriyle tutarlı gibisinden.

Revisions in Manuscript:

↪ **Section 12, Page 23:**
Dropout and batch normalization are applied after each layer.

↪ **Section 13, Page 24:**
Batch normalization and ReLU layers are included after each layer.

Reviewer 2, Comment 5

Comment: *The experiments focus on specific architectures (e.g., MMD-based networks and adversarial networks). The paper notes that "convolutional layers are left out of the scope" in some analyses, which reduces relevance to modern architectures like Transformers or ResNets. Testing a broader range of models would strengthen the claims about generalizability.*

Response: Bu doğrultuda aşağıdaki footnote eklendi:

Revisions in Manuscript:

↪ **Section 7, Page 16:**

Feature extractor networks in common architectures typically consist of several convolutional layers followed by fully connected layers [47]. In domain adaptation applications, it is standard practice to adopt convolutional layer weights from pretrained networks or to train them using source data [54]. Our theoretical analysis focuses on the fully connected layers of the feature extractor, treating the input source and target samples $x^s, x^t \in R^{d_0}$ *as the responses generated at the output of the convolutional network common between the two domains, as illustrated in Figure 3. Notably, our experimental validation in Section 4 employs complete architectures that include convolutional layers trained end-to-end; the consistency of the experimental results with the theoretical predictions suggests that our bounds remain applicable to practical architectures incorporating convolutional components.*